

Class #12 - Inverse of a Matrix

1. Notation: $A^2 =$ _____ (for what A ?)

$A^k =$ _____ where $k = 1, 2, 3, \dots$

2. Example: $A = \begin{bmatrix} 0 & 3 \\ 1 & -1 \end{bmatrix}$, $x = \begin{bmatrix} -5 \\ 3 \end{bmatrix}$

(a) $(Ax)^T$

(b) A^3

(c) xx^T

(d) $x^T x$

(e) $A^T x^T$

(f) $x^T A^T$

3. Def: The $n \times n$ matrix obtained by a *single* elementary row operation to the $n \times n$ identity matrix is called an *elementary matrix*.

Examples: $E_1 = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$

How about $E_4 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, and $E_5 = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$?

4. Note: For any elementary matrix E , the matrix EA is the row equivalent matrix to A via the row operation indicated by E .

5. Def: Let $A \in \mathbf{R}^{n \times n}$. The matrix C such that

$$AC = I = CA$$

is called the *inverse of A* and we write A^{-1} .

If a matrix A has an inverse, we say that A is *invertible* or *nonsingular*.

6. Note:

(a) $C \in$ _____

(b) $A^{-1} \neq \frac{1}{A}$

7. Theorem: Let $A \in \mathbf{R}^{2 \times 2}$ and let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Note: If $ad - bc = 0$, A^{-1} _____.

This number $ad - bc$ determines the _____ of A , so we call this number *the determinant of A* .

Notation: $\det A = ad - bc$ (for 2×2 only!)

8. Example: $A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$. Is A invertible? If so, what is A^{-1} ?

9. Question: For an invertible matrix A , how many inverses of A are there?

Let B and C be both inverses of A .

Then _____ $= I =$ _____, and _____ $= I =$ _____.

Then $B = B$ _____ $= B(\text{_____}) = (\text{_____})C =$ _____ $C = C$.

Hence, $B = C$ and A^{-1} is unique!

Theorem: The inverse of an invertible matrix is unique!

10. Question: If B is the inverse of A , what is the inverse of B ?

Note: _____ $= I =$ _____

11. Theorem: Let A and B be invertible matrices of the size $n \times n$. Then,

(a) $(A^{-1})^{-1} =$ _____

Proof:

(b) $(AB)^{-1} =$ _____

Proof:

(c) A^T is also invertible and $(A^T)^{-1} =$ _____

Proof: Let B be the inverse of A . Then, _____

$I = I^T = (\text{_____})^T = (\text{_____})^T$

Hence, _____ $= I =$ _____.

Therefore, $(A^T)^{-1} =$ _____.